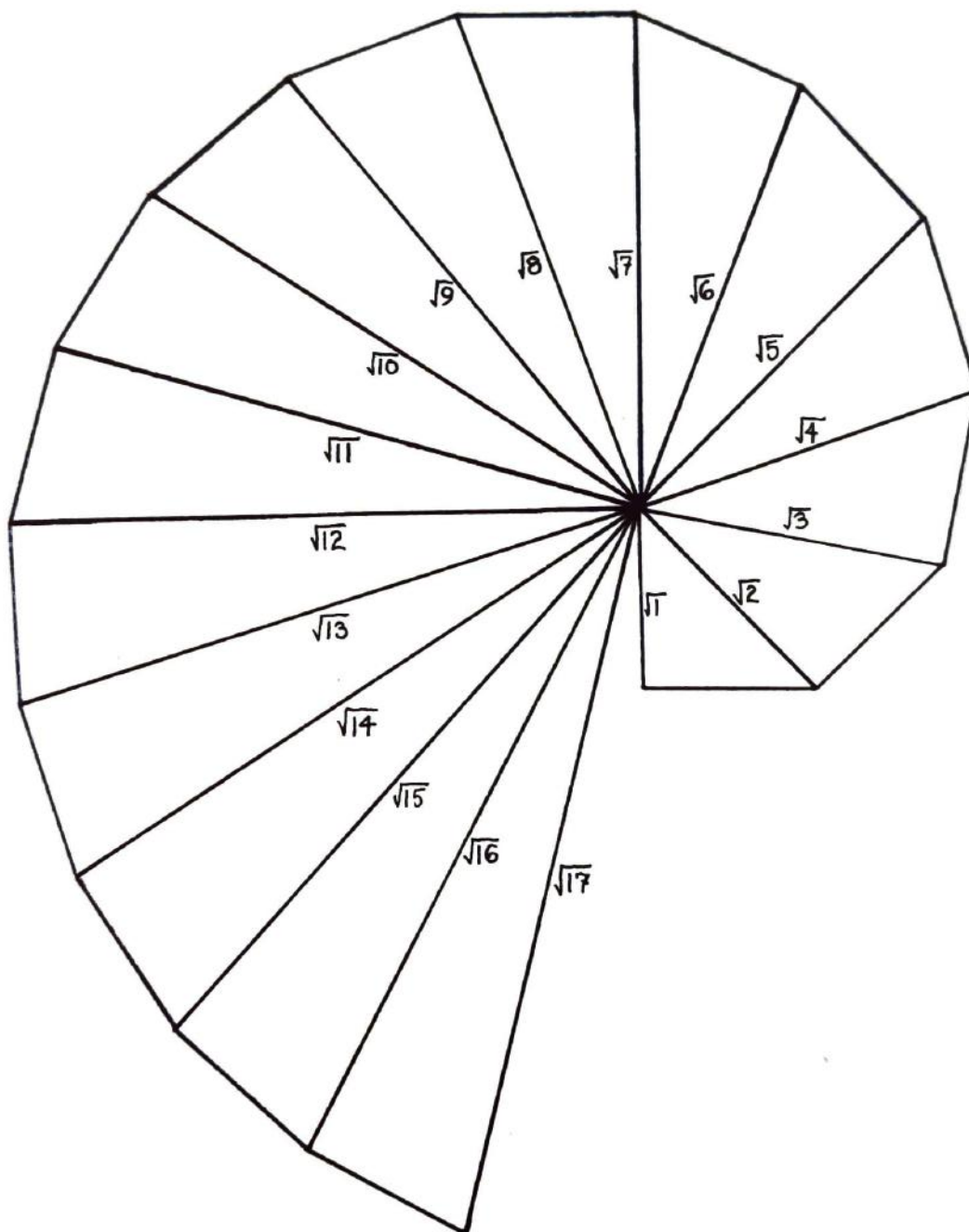


MATH SURVEY



STUYVESANT H.S. 1983

MATH SURVEY

Spring 1983 Vol. LVIII

THE STUYVESANT HIGH SCHOOL

MATHEMATICS PUBLICATION

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CEVA'S THEOREM

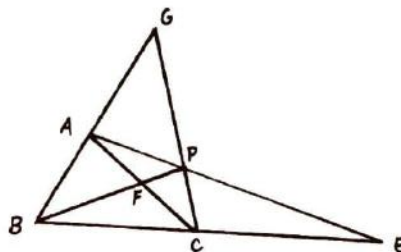
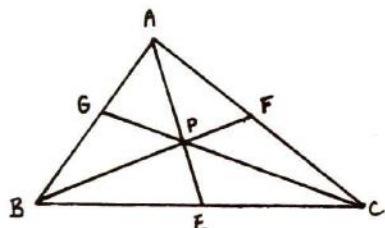
by Bill Rugolsky

Many theorems in geometry depend upon the concurrency of given line segments in a triangle. But because traditional proofs are often awkward, concurrency relationships are generally accepted without proof. With the aid of a theorem first developed by Giovanni Ceva in 1687, much simpler proofs of concurrency relationships can be devised.

Ceva's Theorem:

The three lines or cevians containing the vertices A, B, and C of $\triangle ABC$ and intersecting the opposite sides at points E, F, and G, respectively, are concurrent if and only if

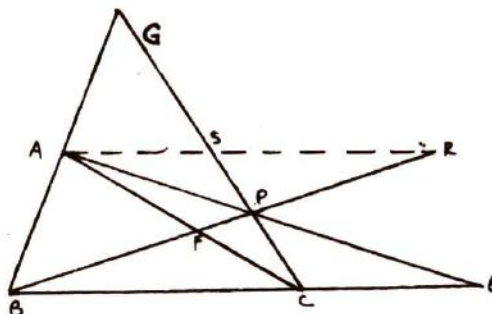
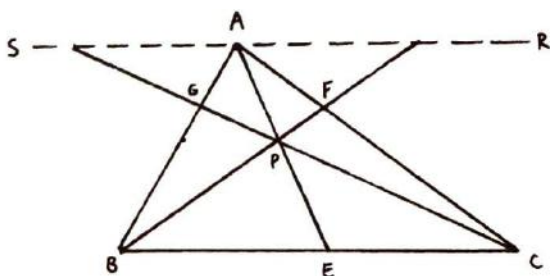
$$\frac{AG}{GB} \cdot \frac{BE}{EC} \cdot \frac{CF}{FA} = 1.$$



Proof:

We note that two cases exist in which the cevians may intersect the sides of the triangle and be concurrent. In each case, the statements presented in the proof hold for both cases and diagrams. Let $\overleftrightarrow{AE}$, $\overleftrightarrow{BF}$, and $\overleftrightarrow{CG}$ intersect at P.

We begin by drawing a line parallel to BC, passing through A, and intersecting $\overleftrightarrow{CP}$ at S and $\overleftrightarrow{BP}$ at R.



From the parallel lines we establish that

$$\triangle AFR \sim \triangle CFB, \text{ or } \frac{AF}{FC} = \frac{AR}{CB} \quad (1)$$

$$\triangle BGC \sim \triangle AGS, \text{ or } \frac{BG}{GA} = \frac{CB}{SA} \quad (2)$$

$$\triangle CEP \sim \triangle SAP, \text{ or } \frac{CE}{SA} = \frac{EP}{AP} \quad (3)$$

$$\triangle BEP \sim \triangle RAP, \text{ or } \frac{BE}{RA} = \frac{EP}{AP} \quad (4)$$

From (3) and (4) we get:

$$\frac{CE}{SA} = \frac{BE}{RA},$$

which can be written as:

$$\frac{CE}{BE} = \frac{SA}{RA} \quad (5)$$

If we multiply (1), (2), and (5) we get:

$$\frac{AF}{FC} \cdot \frac{BG}{GA} \cdot \frac{CE}{BE} = \frac{AR}{CB} \cdot \frac{CB}{SA} \cdot \frac{SA}{RA} = 1$$

By inverting the fractions and shuffling the terms we get

$$\frac{AG}{GB} \cdot \frac{BE}{EC} \cdot \frac{CF}{FA} = 1$$

which is the desired result.

To complete our proof, it is necessary to show that the converse is also true. Namely, if the lines containing the vertices A,B,C intersect the opposite sides at E,F,G, respectively, so that $\frac{AG}{GB} \cdot \frac{BE}{EC} \cdot \frac{CF}{FA} = 1$, then the lines are concurrent.

Proof:

Suppose $\overleftrightarrow{BF}$ and $\overleftrightarrow{AE}$ intersect at P. By drawing line $\overleftrightarrow{PC}$ and labeling its intersection with the line $\overleftrightarrow{AB}$ as G', we can apply part 1 of Ceva's theorem. Thus,

$$\frac{AG'}{G'B} \cdot \frac{BE}{EC} \cdot \frac{CF}{FA} = 1$$

From our hypothesis we know:

$$\frac{AG}{GB} \cdot \frac{BE}{EC} \cdot \frac{CF}{FA} = 1.$$

Therefore: $\frac{AG'}{G'B} = \frac{AG}{GB}$, and the points must coincide, thereby proving concurrency.

For more on Ceva's Theorem and many useful applications, see:
ALFRED S. POSAMENTIER, EXCURSIONS IN ADVANCED EUCLIDEAN GEOMETRY, 1979,
J. Weston Walch Publishers, pp. 24-34.

COMPLIMENTS

OF

THE STUYVESANT HIGH SCHOOL

PARENTS' ASSOCIATION

A HYPERSPACE EXPERIENCE

by Jean Lee

"Oh, no..."

The "CAUTION -Entering Unknown Territory" sign on the control panel flashed on and off furiously. "What are we going to do?" queried Little Jimmy, whirling about in the swivel chair by the panel.

Professor Georg Friedrich Bernhard "Bud" Riemann, who was the head of the cosmic excursion, strolled placidly over to the control panel. "There's nothing to be anxious about," reassured the professor. "Hand me the pen and clipboard on that desk and wake up your sister."

At that moment, a series of numbers appeared on the screen, Professor Riemann jotted down the figures and was consulting a copy of the Universal Reference as Little Jimmy re-entered the main room of the ship with a somewhat groggy Little Janey. "Well," sighed the professor, "I guess we'll be a little late to Professor Maxth's lecture on curved space. We took a wrong left turn a couple of hundred kilometers back, leaving us with an unexpected side trip into the fourth dimension."

"The wha-a-at??" exclaimed a now wide-awake Little Janey.

"The fourth dimension - also known as hyperspace," replied Professor Riemann. "But don't worry, if you're still tired, you can go back and finish your nap. I thought it might have been something worse, but we're safe."

Little Jimmy frowned. "How far off course are we?"

"Not much," said the professor. "It can be fixed in a jiffy."

"Whew! For a moment there I thought we might not make it back in time for the publication of Stuyvesant's 1983 Math Survey," said Little Jimmy, much relieved.

"It doesn't look any different out there in the fourth dimension," mumbled Little Janey, pressing her nose against the glass of the porthole.

"Of course not," said Professor Riemann. "That's because we're three-dimensional (3D) creatures, and our senses can only perceive things in the three dimensions that we are familiar with."

All of this excitement had been getting Little Jimmy a bit restless and hungry, so he reached into a nearby cupboard and withdrew a bag of pretzels. However, he was surprised to find that all three cups which he had placed in the cupboard had disappeared. "Who took all of the cups?"

"Not me," replied the professor and Little Janey simultaneously. Little Jimmy was dumbfounded.

"Perhaps some four-dimensional (4D) being took them," suggested Professor Riemann.

"Huh?"

"Many of the enclosures that we know of in three-space - such as safes, cupboards, and locked rooms without windows - are useless in hyperspace," explained Professor Riemann. "That extra dimension allows a fourth-dimensional creature to reach into the enclosure

and take out anything that we've put in there without unlocking, breaking, or stretching the enclosure."

"I don't see how that works," said a puzzled Little Jimmy.

"We can show this by an analogy from two dimensions into three." The professor proceeded to draw several figures on a piece of paper. "Imagine a two-dimensional creature that lives in the plane represented by this sheet of paper. For him,..."

"...or her," added Little Janey.

"All right we'll call this two-dimensional creature Q in order to avoid any assumptions." Little Janey nodded in agreement. "Now for Q, any non-degenerate, simple closed curve in the plane could serve as an enclosure in which to store and lock up goods (see fig. 1). But we, being 3D creatures, can use this third dimension that Q cannot perceive to take things in and out of Q's locked enclosure. We would have no need to unlock, break, or stretch Q's enclosure in order to do this (see fig. 2). To Q, it would appear as if the object that was stored in the enclosure had magically vanished and then re-appeared. This would correspond to a 3D being's removal and replacement of the object from Q's plane," explained the professor.

A two-dimensional creature

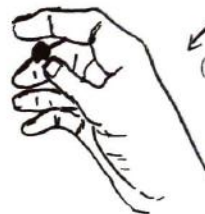
Q



An enclosure or safe in two dimensions

Fig. 1

Q



Hand
(from third dimension)

Fig. 2

Little Jimmy offered the professor a pretzel.

"No, thank you," he declined. "But in hyperspace all of those pretzels could be unknotted without stretching or breaking any of them."

"And this is because of the extra dimension that the fourth dimension provides to make the unknotting possible," suggested Little Jimmy.

"Exactly. There are all sorts of other nice phenomena that can occur in the fourth dimension," continued Professor Riemann.

"Like what?" chorused Little Jimmy and Little Janey.

"Well, your innards would be exposed in hyperspace to any 4D creature!"

Little Janey turned a pale shade of green.

"I guess it would be in the same way in which we - as beings of three-space - could easily look into Q's insides," said Little Jimmy. "Our third dimension of height lets us look over into Q's 2D world of the plane."

"That's right," said Professor Riemann. "You could also revolve objects about a plane in the fourth dimension." The professor drew some more figures on the paper. "If we wanted to get quadrilateral A to look like quadrilateral B - in the same plane, of course - what would we do?" (see fig 3)

"I'd use a mirror and take the reflection of quadrilateral A," remarked Little Jimmy.

"That's cheating!" said Little Janey. "It's supposed to be taken through a 180° rotation in three-space!"

"Very good. Similarly, a 180° rotation through four-space can be used to get solid A to look like solid B," (see fig 4) added Professor Riemann.

"And it would have to be a 180° rotation about a plane in the latter case," said Little Jimmy, "since objects in n -space can only be revolved about a space of order $n-2$, where $n \geq 2$. But I still think using a mirror would be easier."

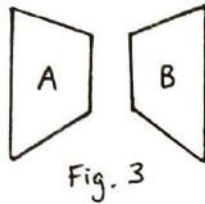


Fig. 3

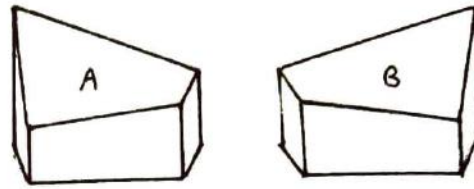


Fig. 4

"But how do we know that hyperspace exists? It's so difficult to visualize!" whined Little Janey.

"Many of the equations of mathematics and physics indicate the existence of n dimensions - certainly the basic equations of analytic geometry can be extended to n dimensions," began the professor. "As for visualization, there's always something called the imagination," said the professor, pointing to his temple.

"But what can you do with the fourth dimension?" asked Little Jimmy.

"Why, Jim!" exclaimed the professor. "If you had just used your head a little, you would have known that the study of the fourth and upper dimensions is very useful when graphing curves, where each dimension represents each of the independent variables affecting the curve. This would be a big help to those involved with the behavioral sciences or catastrophe theory. Also, many spiritualists conjecture that ghosts are four dimensional creatures, since that would explain some of the things that they have been claimed to do."

"The fourth dimension might also be an explanation for the formation of isomerisms in chemistry," piped Little Janey.

At that moment, the sign on the control panel stopped flashing. Little Jimmy sat in quiet contemplation of the fourth dimension, captivated by its strange yet logical implications.

"Look what I found!" shouted Little Janey. She held open the door of the cupboard to reveal three cups on the top shelf. Little Jimmy glanced at the professor, and they smiled.

*

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If you, like Little Jimmy and Little Janey, have also found the fourth dimension to have captured your imagination, you may want to read Edwin Abbott's Flatland and H. P. Manning's The Fourth Dimension Simply Explained for a more detailed discussion of the topic.

THE DELIAN CUBE- DOUBLING METHOD

by Tzea-Ning Chen

The name Delian, according to an account given by the mathematician and historian Eutocius (sixth century A. D.), goes back to an old legend according to which the Delphic oracle in one of its utterances demanded that the Delian altar block be doubled. In mathematical terms, the problem is: to construct the edge of a cube that doubles the volume of a given cube. (Note: this problem cannot be solved with a compass and a straightedge)

The solution goes as follows:

Let k = the length of the edge of the given cube.

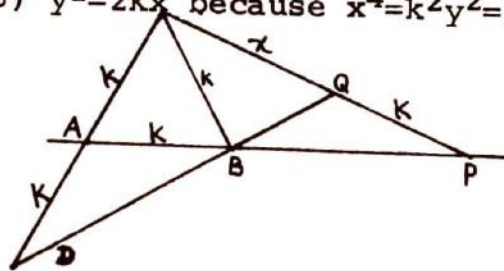
Let x = the length of the edge of the cube that is doubled.

Thus: (1) $x^3 = 2k^3$;

which has the same solutions as:

(2) $x^2 = ky$, and (3) $y^2 = 2kx$ because $x^4 = k^2 y^2 = 2k^3 x$ and $x \neq 0$, so $x^3 = 2k^3$.

Construction:



1. Draw equilateral ABC with side k .
 2. Extend CA by $AD = k$.
 3. Draw DB.
 4. Mark off the distance k on the paper edge strip.
 5. Place the paper strip such that the edge passes through C and the endpoints of the marked-off distance fall through points P and Q of the extensions of AB and DB.
- $\therefore CQ = x = k\sqrt[3]{2}$

Proof:

- (a) Let $CQ = x$, $BP = y$; then, according to the Leg Transversal Theorem used in figure CABP:

$$(x+k)^2 - k^2 = y(k+y)$$

or

$$(4) x^2 + 2kx = y^2 + ky$$

- (b) By Menelaus Theorem, on triangle ACP with transversal DBQ:

$$AD \cdot CQ \cdot BP = PQ \cdot AB \cdot CD$$

or

$$(5) xy = 2k^3$$

- (c) Therefore, the solutions of equations (4) and (5) in the proof would be the same as those of equations (2) and (3) in the solution.

BOLD PLAY

by Kenneth Baron

Probability and mathematics have surprisingly momentous applications to games of chance. We will discuss one such technique known as Bold Play.

Consider the case where player K enters a casino with \$100. He will play at the casino until he either reaches his goal of \$500 or he goes broke. He plays in a game where his probability of winning is $1/2$. (You would be hard pressed to find these odds at a real casino.) The amount he bets on each play is inconsequential to his chances of reaching his goal. His chance of reaching that goal of \$500 is his present capital = $\frac{\$100}{\$500} = 0.2$ (Since he has a .8 chance of finishing with \$0 and a .2 chance of finishing with \$500, his expected value of money at the end is $.8(0) + .2(500) = \$100$. Thus these probabilities are, in effect, just a consequence of the game being fair, since his expected value equals his initial value.) When Player K needs \$20000 and starts with \$100 his chance is .005.

Let's take a real situation where Player K goes to play the Roulette Wheel. Here Player K can bet on either the color red or black. There are 18 slots of each color. Also, there are 2 slots with the color green. If the Wheel lands on green the house automatically wins. Thus, Player K has an $\frac{18}{38}$ chance of winning on any given turn. Let b be the

amount Player K bets on each turn; c be the capital Player K starts with; g be the goal of Player K; p be the probability that Player K wins on any given turn; $q=(1-p)$ be the probability that Player K loses on any given turn. Then his chance of reaching his goal is given by the formula:

$$\frac{\left(\frac{q}{p}\right)^{\frac{c}{b}} - 1}{\left(\frac{q}{p}\right)^{\frac{g}{b}} - 1}. \text{ If } q > p, c \gg b, \text{ and } g \gg b, \text{ then } \frac{\left(\frac{q}{p}\right)^{\frac{c}{b}} - 1}{\left(\frac{q}{p}\right)^{\frac{g}{b}} - 1} \approx \frac{\left(\frac{q}{p}\right)^{\frac{c}{b}}}{\left(\frac{q}{p}\right)^{\frac{g}{b}}} = \left(\frac{q}{p}\right)^{\frac{c-g}{b}} = \left(\frac{p}{q}\right)^{\frac{g-c}{b}}$$

Though $\frac{18}{38}$ is very close to $\frac{1}{2}$, the difference becomes gigantic when they are multiplied over many times. For example, Player K's chance of winning \$500 from a starting amount of \$100 with \$1 bets is 5×10^{-9} . Also, if Player K tries to win \$20,000 from \$100, his chance is 3×10^{-11} . The difference between the chance of winning with a $\frac{1}{2}$ probability and a $\frac{18}{38}$ probability is substantial.

These probabilities would be very discouraging to any gambler. Bold Play helps to increase a gambler's chance of reaching a goal where he has less than a $1/2$ chance of winning on any given turn. In Bold Play,

if the capital (c), the player started with is less than half of his goal (g), then he bets all of c . However if c is greater than half of g , then he bets $g-c$. This process is continued until the player either reaches his goal or goes broke.

To see how Bold Play works, let's try an example. Once again consider the case where Player K has \$500 a goal and is starting with \$100. On

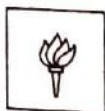
his first try he bets all \$100. If he bet less than \$100 his chance of reaching his goal would lessen because of the above formula's compounding effect on probabilities less than $\frac{1}{2}$. If the player wins on the first turn, then he bets all \$200 for the reason above. If he wins on this turn, then he needs only \$100 to reach his goal of \$500. Thus he will only bet \$100. If he bets all \$400, then he would needlessly be risking \$300. If he wins, he will stop. But if he loses, he still has \$300 to play with. This process is continued until he is either broke or successful. Bold Play minimizes the number of turns and thus maximizes the chance of winning.

How much does Bold Play increase the player's chances? Well, Player K's chance of winning \$500 from \$100 is now .18. Compare this to the previous figures of .2 and 5×10^{-22} . Also if Player K uses Bold Play when he has \$100 and needs \$20,000, his chance is .003 as compared with .005 and 3×10^{-911} .

As shown above Bold Play can make a considerable difference. Using simple concepts and techniques Bold Play can increase tremendously someone's chance of winning and reaching his goal.

<u>Capital</u>	<u>Goal</u>	<u>Stakes</u>	<u>Probability</u>	<u>Final Probability</u>
\$100	\$500	\$1	.5	.2
\$100	\$20,000	\$1	.5	.005
\$100	\$500	\$1	18/38	5×10^{-22}
\$100	\$20,000	\$1	18/38	3×10^{-911}
\$100	\$500	BOLD PLAY	18/38	.18
\$100	\$20,000	BOLD PLAY	18/38	.003

Article based on lecture at the University of Chicago: Summer Mathematics Program of 1982.



COMPLIMENTS OF:
NEW YORK UNIVERSITY
COURANT INSTITUTE OF
MATHEMATICAL SCIENCES

PYTHAGOREAN TRIPLES

by Delilah Lopez

Anyone who has taken algebra or geometry recalls the Pythagorean theorem and its relationship with the right triangle. Some can even remember some Pythagorean triples. But, have you ever thought of how these triples are found? Many mathematicians have found this problem quite fascinating. Even before the mighty Diophantus (250 AD) and Euclid (300 BC), the ancient Babylonians worked on this brainteaser. There is a cuneiform tablet (1500 BC) at an archaeological collection at Columbia University with the triples 3,4,5 to 4961, 6480, 8161. This incredible problem has been solved by many greats such as Dio- phantus and Fermat.

Here's the solution: (Based on Fermat's)

If you are given $x^2 + y^2 = z^2$ where x , y , and z are positive integers, then x/d , y/d , z/d is also a triple since the d^2 cancels out in $\frac{x^2}{d^2} + \frac{y^2}{d^2} = \frac{z^2}{d^2}$. If d is the greatest common divisor, then

$\frac{x}{d}$, $\frac{y}{d}$ and $\frac{z}{d}$ have no common divisor other than 1 making it a primitive

P. triple (a,b,c) (eg. 3,4,5). Every triple can be reduced to a primitive by dividing by the greatest divisor ($6,8,10 \div 2=3,4,5$), and visa versa.

Being a primitive triple, (a,b,c) cannot be reduced, therefore all three numbers cannot be even. On the other hand they can't all be odd since odd+odd $\neq$ odd. It is then necessary for one of the numbers to be even. But which one? If we take the integers modulo 4, it can easily be seen that c must be odd. The square of an odd number $(2n+1)$, $(4n^2+4n+1)$, is one more than a multiple of 4. If a and b are odd, implying that c is even, then a^2+b^2 is 2 more than a multiple of 4. But since c is even, $c=2k$, so $c^2=4k^2$ is a multiple of 4. This is a contradiction, so a and b are of opposite parities (one odd and one even), and c is odd. By doing this, we can change $a^2+b^2=c^2$ to $a^2=c^2-b^2$ and that to $a^2=(c+b)(c-b)$. a , $c+b$ and $c-b$ are all even integers (odd-odd=even; odd+odd=even). Replacing these numbers with u,v , and w , such that $a=2u$; $c+b=2v$; and $c-b=2w$, we get $(2u)^2=(2v)(2w)$; $u^2=vw$, v and w are relatively prime. This can be proven, because any number dividing them would also divide $v+w=1/2(c+b)+1/2(c-b)=c$ and $v-w=1/2(c+b)-1/2(c-b)=b$ and b and c are relatively prime.

From this, we proceed to the main step of the proof. The only way that the product of two relatively prime numbers can equal a square is if each are squares. (Since their product is a square, all primes in its prime factorization must appear to an even power. Since the two numbers have no common prime factors, all primes in the prime factorization of each must appear to an even power, implying that v and w are squares.)

Following this, we can then substitute v with p^2 and w with q^2 getting $c=v+w=p^2+q^2$ and $b=v-w=p^2-q^2$. In order for b to be positive and odd $p > q$ and p and q must be of opposite parities. To express a in terms of p and q

we merely substitute:

$$a^2 = c^2 - b^2$$

$$a^2 = (p^2 + q^2)^2 - (p^2 - q^2)^2$$

$$a^2 = p^4 + 2p^2q^2 + q^4 - p^4 + 2p^2q^2 - q^4$$

$$a^2 = 4p^2q^2$$

$$a = 2pq.$$

Therefore $(2pq)^2 + (p^2 - q^2)^2 = (p^2 + q^2)^2$. This formula works to find any primitive pythagorean triples if 1) p and q are relatively prime
2) $p > q$ and 3) p and q are of opposite parities.

Example : If $p=2$ and $q=1$ $(2 \cdot 2 \cdot 1)^2 + (4-1)^2 = (4+1)^2$
 $4^2 + 3^2 = 5^2$

Problem : Find p and q for the triple (4961, 6480, 8161). Ans. $p=81, q=40$

Math Team

Every morning, a group of extremely dedicated students assembles at 8:00 in room 517. These bleary-eyed students sacrifice their sleep to attend Math Team classes supervised by Coach Dr. Richard Rothenberg and conducted by Captains Larry Buxbaum and David Zuckerman.

Stuyvesant did quite well on this year's Annual High School Mathematics Examination (MAA). Our team score of 398 is the highest in New York City, and likely to be the highest in the country, thanks to the scores of 145, 130, and 123 (out of a possible 150) by Zuckerman, Kevin Buehler, and Buxbaum, respectively. Zuckerman's score was tied for highest in the country. For the first time, this year there was another contest after the MAA to determine who will take the United States Math Olympiad. Thirty Stuyvesant students scored at least 95 on the MAA to qualify to take this test, called the American Invitational Mathematics Examination (AIME).

In the New York Math League competition this year, Stuyvesant is doing especially well. As of this writing, we are in first place and Zuckerman has the only perfect score.

Last term, our First Senior Team finished third, the first time in 17 terms we did not win. It consisted of Buehler, Buxbaum, Jay Sipelstein, Carl Weiner, and Zuckerman. The Second Senior Team, comprised of Kenneth Baron, Iuval Clejan, Anna Franco, Nir Hachohen, Naiyer Imam, David Lam, and Darrell Tornay also did well. In the Junior Division, our First and Second Teams finished second and third, respectively. Captain Ted Poon had the highest score in the city and Jean Lee tied for second. The First Team consisted of Michael Friedman, Andrew Lee, Poon, Dana Randall, and William Rugolsky, and the Second Team of Mark Basch, Joel Hasen, David Kadmon, Lee, and Sui-Ming Louie.

Math Team has had record attendance this year. If you are interested in learning to solve a variety of problems, and having a good time in the process, come and join us!

THE SOLUTION TO A CUBIC EQUATION

by Jay Sipelstein

The term "solution to an equation" refers to a method that will enable one to determine all of the roots of an equation, whether real or complex, either exactly or to any desired degree of accuracy. The solution to the cubic equation was first published by the Italian algebraist Gerolamo Cardano (1501-1576). In this article, I will present a derivation of Cardano's formulas.

Given an arbitrary cubic equation, there is no loss of generality in dividing its coefficients by the coefficient of x^3 , so we may let the general cubic take the form:

$$f(x) = x^3 + ax^2 + bx + c = 0. \quad (1)$$

The solution to (1) will be greatly facilitated by the introduction of a new unknown that will remove the ax^2 term of (1). If we let x_1, x_2, x_3 be the roots of (1), we have:

$$(x-x_1)(x-x_2)(x-x_3) = 0,$$

or, by multiplication,

$$x^3 - (x_1+x_2+x_3)x^2 + (x_1x_2+x_2x_3+x_1x_3)x - x_1x_2x_3 = 0.$$

By equating coefficients we see that $a = -(x_1+x_2+x_3)$. The removal of the ax^2 term of (1) is equivalent to setting a equal to 0. A new cubic in y , with every root equal to a corresponding root of (1) plus the average of the three roots of (1), will have this desired property. This statement can be represented by the equation:

$$x = y - (a/3). \quad (2)$$

If this substitution is made, (1) is transformed into the related equation:

$$y^3 + py + q = 0 \quad (3)$$

where $p = b - (a^2/3)$ and $q = c - (ba/3) + (2a^3/27)$.

The reader can verify this by much arduous algebra if (s)he is in a masochistic mood.

The solution to (3) is accomplished by introducing the unknowns m and n such that:

$$y^3 + py + q = y^3 + m^3 + n^3 - 3mny \quad (4)$$

or, by equating coefficients and cubing the second relation:

$$m^3 + n^3 = q \quad \text{and} \quad m^3 n^3 = -p^3/27.$$

If we let m^3 and n^3 be roots of a quadratic in z , we have:

$$z^2 - (m^3 + n^3)z + m^3 n^3 = 0$$

$$z^2 - qz - (p^3/27) = 0$$

Using the quadratic formula and taking cube roots:

$$m = (q/2 + (q^2/4 + p^3/27)^{1/2})^{1/3} \quad \text{and} \quad n = (q/2 - (q^2/4 + p^3/27)^{1/2})^{1/3} \quad (5)$$

The actual roots of (3) can be found by the use of the algebraic identity:

$$y^3 + m^3 + n^3 - 3mny = (m+n+y)(a^2 + b^2 + y^2 - ab - ay - by) \quad (6)$$

Combining (3), (4), and (6) we have:

$$m+n+y=0 \quad \text{and} \quad y^2 - (a+b)y + a^2 + b^2 - ab = 0.$$

By letting y_1, y_2, y_3 be the roots of (2) and solving the above equations for y , we find that

$$y_1 = -m-n, \quad y_2 = \frac{m+n}{2} + \frac{(m-n)\sqrt{3}}{2}i, \quad \text{and} \quad y_3 = \frac{m+n}{2} - \frac{(m-n)\sqrt{3}}{2}i \quad (7)$$

By substituting these values into (2), and solving for x , we arrive at the three roots of (1). Since there are no restrictions placed on the coefficients in any of the operations involved in Cardano's formulas, the coefficients of (1) can have any real or complex value, and thus we have a method by which any cubic equation can be solved exactly!

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LINEAR ALGEBRA

by David Zuckerman

Linear algebra is an important part of mathematics. We will use some of the fundamentals of linear algebra to derive a seemingly unrelated and interesting result.

X is called a linear space over a field K of scalars if and only if, for arbitrary x, y, z in X and h, k in K , $x+y$ and kx are defined and:

- 1) $x+(y+z) = (x+y)+z$
- 2) $x+y = y+x$
- 3) There exists a 0 in X such that $x+0 = x$
- 4) There exists a $(-x)$ in X such that $x+(-x) = 0$
- 5) $k(hx) = (kh)x$
- 6) $k(x+y) = kx+ky$ and $(k+h)x = kx+hx$
- 7) $1x = x$, where 1 is the multiplicative identity of K .

Definition: The vectors $x_1, x_2, \dots, x_n$ in X are linearly dependent if and only if there exist $k_1, k_2, \dots, k_n$ in K such that

$$k_1x_1+k_2x_2+\dots+k_nx_n = 0.$$

Definition: The vectors $x_1, x_2, \dots, x_n$ in X span the space X if and only if any x in X can be written as a linear combination of $x_1, x_2, \dots, x_n$, that is, there exists $k_1, k_2, \dots, k_n$ in K such that $x=k_1x_1+k_2x_2+\dots+k_nx_n$.

Theorem 1: If $x_1, x_2, \dots, x_n$ span X and $y_1, y_2, \dots, y_j$ in X are linearly independent (not linearly dependent), then $j \leq n$.

Proof: Since $x_1, x_2, \dots, x_n$ span X and $y_1 \in X$, there exist $k_1, k_2, \dots, k_n$ in K such that

$$y_1 = k_1x_1 + k_2x_2 + \dots + k_nx_n.$$

Since $y_1 \neq 0$ (or else $ky_1 + 0y_2 + \dots + 0y_n = 0$ and the y_i 's are not linearly independent) at least one of the k_i 's does not equal zero. We may assume without loss of generality that this is k_1 . Thus,

$$x_1 = \frac{1}{k_1}y_1 - \frac{k_2}{k_1}x_2 - \frac{k_3}{k_1}x_3 - \dots - \frac{k_n}{k_1}x_n.$$

Since $x_1, x_2, \dots, x_n$ span X , for any vector x there exist scalars $h_1, h_2, \dots, h_n$ such that

$$\begin{aligned} x &= h_1x_1 + h_2x_2 + \dots + h_nx_n \\ &= h_1\left(\frac{1}{k_1}y_1 - \frac{k_2}{k_1}x_2 - \frac{k_3}{k_1}x_3 - \dots - \frac{k_n}{k_1}x_n\right) + h_2x_2 + \dots + h_nx_n \\ &= \frac{h_1}{k_1}y_1 + \left(h_2 - \frac{h_1k_2}{k_1}\right)x_2 + \dots + \left(h_n - \frac{h_1k_n}{k_1}\right)x_n \end{aligned}$$

Thus any vector can be written as a linear combination of $y_1, x_2, x_3, \dots, x_n$, so these vectors span X . By expressing y_2 as $y_2 = k_1' y_1 + k_2' x_2 + \dots + k_n' x_n$, and noting that some $k_i \neq 0, i \geq 2$ (if not, then $y_2 = k_1' y_1$ or $y_2 - k_1' y_1 = 0$ and the y_i 's are not linearly independent), we can use the same argument to show that $y_1, y_2, x_3, x_4, x_5, \dots, x_n$ span X . If we suppose that $j > n$, then $y_1, y_2, y_3, \dots, y_n$ span X , so $y_{n+1} = h_1' y_1 + h_2' y_2 + \dots + h_n' y_n$, and the y_i 's are not linearly independent. This is a contradiction; hence, $j \leq n$.

Definition: $x_1, x_2, x_3, \dots, x_n$ constitute a basis for X if and only if they are linearly independent and span X .

Definition: The dimension of X (abbreviated $\dim X$) is n if $x_1, x_2, \dots, x_n$ form a basis for X .

This is well defined, for if $y_1, y_2, \dots, y_m$ form a basis, then the y_i 's are linearly independent and the x_i 's span X . By theorem 1, $m \leq n$. Similarly, the x_i 's are also linearly independent, and the y_i 's span X , so $n \leq m$. Thus $n = m$ by the Squeeze Theorem.

Definition: A function T is said to be linear iff $T(x+y) = T(x) + T(y)$ and $T(kx) = kT(x)$. We will abbreviate $T(x)$ as Tx , and let T map X into U .

Definition: N_T , the null space of T , is the set of elements in X that T maps into 0 .

Definition: R_T , the range of T , is the set of elements u in U for which there exists an x in X such that $T(x) = u$.

Theorem 2: Suppose $\dim X = \dim U = n$, and $N_T = \{0\}$, and let $x_1, x_2, \dots, x_n$ be a basis for X . Then $Tx_1, Tx_2, \dots, Tx_n$ form a basis for U , and $R_T = U$.

Proof: If $Tx_1, Tx_2, \dots, Tx_n$ were not linearly independent, then

$$k_1 Tx_1 + k_2 Tx_2 + \dots + k_n Tx_n = 0, \text{ or}$$

$$T(k_1 x_1) + T(k_2 x_2) + \dots + T(k_n x_n) = 0$$

$$T(k_1 x_1 + k_2 x_2 + \dots + k_n x_n) = 0$$

Since $N_T = \{0\}$, the only element T maps into 0 is 0 , so $k_1 x_1 + k_2 x_2 + \dots + k_n x_n = 0$. This contradicts our assumption of the x_i 's being linearly independent.

To show that $Tx_1, \dots, Tx_n$ span U , let u be an arbitrary element in U . Since any $n+1$ vectors are linearly dependent,

$$h_1 Tx_1 + h_2 Tx_2 + \dots + h_n Tx_n + h_{n+1} u = 0.$$

If $h_{n+1}=0$, then $Tx_1, \dots, Tx_n$ would be linearly dependent. Thus $h_{n+1} \neq 0$, so

$$u = -\frac{h_1}{h_{n+1}}Tx_1 - \frac{h_2}{h_{n+1}}Tx_2 - \dots - \frac{h_n}{h_{n+1}}Tx_n.$$

Thus $Tx_1, \dots, Tx_n$ span U .

To show $R_T = U$, suppose we wish to write u as $u=Tx$, $x \in X$. First we write u as:

$$\begin{aligned} u &= h_1'Tx_1 + h_2'Tx_2 + \dots + h_n'Tx_n = T(h_1'x_1) + T(h_2'x_2) + \dots + T(h_n'x_n) = \\ &= T(h_1'x_1 + h_2'x_2 + \dots + h_n'x_n). \end{aligned}$$

Thus we can write any element u as Tx , so $R_T = U$.

We are now ready for our interesting result:

There exists a polynomial of degree $< n$ which passes through any n points in a plane.

Proof: Let X be the set of polynomials p of degree $< n$, and

U be the set of vectors in $\mathbb{R}^n$, $(u_1, u_2, \dots, u_n)$.

Let $T(p) = (p(s_1), p(s_2), \dots, p(s_n))$ for arbitrary $s_1, s_2, \dots, s_n$, $s_i \neq s_j$ for $i \neq j$. If we define the sum of two polynomials as $(p+q)x = p(x) + q(x)$ for all x , and $(kp)x = k(p(x))$ for all x , then T is linear. Also $N_T = \{0\}$, for if $T(p) = (0, 0, \dots, 0)$ for some p , then $p(s_1) = p(s_2) = \dots = p(s_n) = 0$, or p has n roots. But since p has degree $< n$, this is possible only if $p(x) = 0$ for all x . Also $\dim X = \dim U = n$, since p can be uniquely defined by its n coefficients $a_0, a_1, \dots, a_{n-1}$ if $p = a_0x^{n-1} + a_1x^{n-2} + \dots + a_{n-1}$. Thus Theorem 2 applies, and $R_T = U$, or for any other u in U , there is some p in X such that $(p(s_1), p(s_2), \dots, p(s_n)) = (u_1, u_2, \dots, u_n)$, or $u_i = p(s_i)$ for $i = 1, 2, \dots, n$, for arbitrary u_i and s_i . Thus there is a polynomial of degree $< n$ passing through $(s_1, u_1), (s_2, u_2), \dots, (s_n, u_n)$ for any s_i, u_i .

Q. E. D.

COMPLIMENTS

OF

THE STUYVESANT HIGH SCHOOL

ADULT CENTER

DIOPHANTINE EQUATIONS

by Larry Buxbaum

Little Zoltan was bored in math class. He flipped through the pages of the now defunct Mathematics Student. He examined problem #557 since the sum of its digits is 17. It read:

#557 Find three distinct integers a, b , and r such that a hexagon, three of whose sides are of measure a while the other three are of measure b , can be inscribed in a circle of radius r .

(Proposer : Professor Tom Butts)

Zoltan scribbled some diagrams and attempted to put the problem in the form GIVEN, FIND, SOLUTION but couldn't.

Suddenly a flash of light appeared. An ancient bearded and robed figure materialized and floated a few inches above the ground in front of the stupified student.

"Hi there! How's it coming?" remarked the robed figure to Zoltan.

"Who are you?" retorted the puzzled student.

"I am Diophantes of Alexandria. (The usual patron mathematician of Math Survey was off on an excursion to the Fourth Dimension at the time.) Did you notice that that problem you are trying to solve is one of my favorite types of problems?" the Mathematician inquired.

"I don't understand."

"Here look at this diagram. If we take two representative sides of length a and b , they are inscribed in an arc of 120° since three pairs add up to the entire circle. The angle between the two radii is therefore 120° and the angle between the two sides of the hexagon is also 120° by elementary geometry. Now use the Law of Cosines on $\triangle ADC$ and $\triangle ABC$.

$$(1) \quad (AC)^2 = a^2 + b^2 - 2ab\cos 120^\circ = a^2 + b^2 + ab$$

$$(2) \quad (AC)^2 = r^2 + r^2 - 2r^2\cos 120^\circ = 3r^2$$

"These yield:

$$(3) \quad a^2 + ab + b^2 = 3r^2$$

which is one of my Diophantine equations!"

"Wow! I hadn't noticed that. Thanks alot," Zoltan said in amazement. "Now all I have to do is solve that simple quadratic in three variables...What? Where am I supposed to go from here?"

"Don't fret young man. Why not solve for a using the quadratic formula," suggested Diophantus.

$$(4) \quad a = \frac{-b \pm \sqrt{b^2 - 4(b^2 - 3r^2)}}{2} = \frac{-b \pm \sqrt{12r^2 - 3b^2}}{2}$$

We must reject the negative value since a and b are both positive integers."

Thus:

$$(5) \quad a = \frac{-b + \sqrt{12r^2 - 3b^2}}{2}$$

"Now I see, the discriminant must be a perfect square for a to be an integer. Let us call

$$(6) \quad 12r^2 - 3b^2 = n^2, \text{ for some positive integer } n$$

$$(7) \quad 3(4r^2 - b^2) = n^2$$

Since 3 is a factor of the left it is a factor of the right. Let $n = 3k$, for some integer k.

$$(8) \quad 3(4r^2 - b^2) = n^2 = 9k^2$$

$$(9) \quad 4r^2 - b^2 = 3k^2$$

$$(10) \quad (2r + b)(2r - b) = 3k^2$$

We can now take the three cases of factoring $3k^2$ and matching factors with the factors on the left side."

Case I

$$(11) \quad 2r + b = 3k$$

$$(12) \quad 2r - b = k$$

$$(13) \quad 4r = 4k \text{ (by adding)}$$

$$(14) \quad r = k$$

$$(15) \text{ and, } b = k \text{ [Subst. (14) into (11)]}$$

This yields $a = b = r$

(The trivial case)

Case II

$$(16) \quad 2r + b = 3k^2$$

$$(17) \quad 2r - b = 1$$

$$(18) \quad 4r = 3k^2 + 1 \text{ (addition)}$$

$$(19) \quad r = (3k^2 + 1)/4$$

$$(20) \quad 2b = 3k^2 - 1$$

(by subtracting (16) & (17))

$$(21) \quad b = (3k^2 - 1)/2$$

$$(22) \quad a = (-b + \sqrt{12r^2 - 3b^2})/2$$

(substitution into (5))

$$(23) \quad a = (1 + 6k - 3k^2)/4$$

Case III

$$(24) \quad 2r + b = k^2$$

$$(25) \quad 2r - b = 3$$

$$(26) \quad 4r = 3 + k^2$$

$$(27) \quad r = (3 + k^2)/4$$

$$(28) \quad 2b = k^2 - 3$$

$$(29) \quad b = (k^2 - 3)/2$$

$$(30) \quad a = \frac{3 + 6k - k^2}{4}$$

(substitution into (5))

"It must be noted that if k is taken to be the product xy we will have several other cases which will be special instances of these three. Hence we may restrict ourselves without loss of generality," the mathematician commented.

"Case I clearly does not fit the conditions. In cases II and III we must clear fractions since a, b, and r are integers. We can multiply a, b, and r by 4 since this corresponds to a scale change in the diagram. Also,

$$(31) \quad (4a)^2 + (4a)(4b) + (4b)^2 = 3(4r)^2 \text{ yields}$$

$$(32) \quad a^2 + ab + b^2 = 3r^2 \text{ when reduced.}$$

Thus far our solutions are

$(a,b,r) = (1 + 6k - 3k^2, 6k^2 - 2, 3k^2 + 1)$ and
 $(3 + 6k - k^2, 2k^2 - 6, 3 + k^2)$, or any integral multiple of these."

Diophantus was impressed, but not too impressed. "What about the solution (13,46,31)? That doesn't fit your solutions?"

"Oh boy!" Zoltan checked it out and the great master of integers was correct.

"Sorry, I have to run. The spirit world calls. Ectoplasm is expensive these days, and you know how expensive these long distance rates are. Here's my model solution for the rest of the problem. Try it yourself. If you can't figure out your error then peak. Bye-bye!"

"Aha! Why not substitute $k = \frac{p}{q}$ and then clear fractions again,"

little Zoltan deduced just as the mathematician vanished.

$$(33) \quad (1 + 6k - 3k^2, 6k^2 - 2, 3k^2 + 1)$$

$$(34) \quad (1 + \frac{6p}{q} - \frac{3p^2}{q^2}, \frac{6p^2}{q^2} - 2, \frac{3p^2}{q^2} + 1)$$

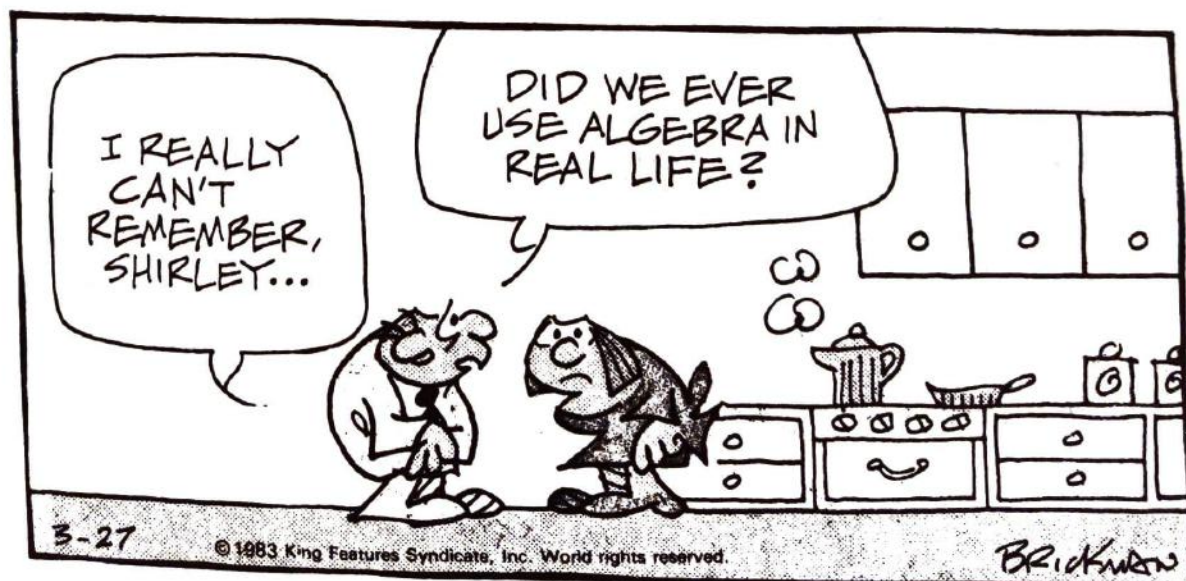
$$(35) \quad (q^2 + 6pq - 3p^2, 6p^2 - 2q^2, 3p^2 + q^2) \quad (\text{clearing fractions})$$

$$(36) \quad (3 + 6k - k^2, 2k^2 - 6, k^2 + 3) \quad \text{similarly yields:}$$

$$(37) \quad (3q^2 + 6pq - q^2, 2p^2 - 6q^2, p^2 + 3q^2)$$

"Now if $p = 3$ and $q = 2$ we get (13,46,31) from the first parametric solution! We set the proper restrictions upon p and q such that a, b , and r are positive. These restrictions can be easily deduced by setting a, b , and r greater than or equal to 0." Zoltan had now completed the problem.

Zoltan awoke with a start. He had a scribbled mess on his paper. Had Diophantus really been there? No one else seemed to have noticed. It had seemed so real. As he left the room, a small piece of parchment with mathematical symbols and ancient Greek writing fell to the floor and was kicked under the teacher's desk. Zoltan stumbled out of the room with a strange feeling.



3-27

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BRICKMAN

Regents Refresher

Quadratic Formula: If $ax^2+bx+c=0$, then $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$, where $d=b^2-4ac$ is the discriminant.

If $d>0$, then the roots are real and distinct.

If $d=0$, then the roots are real and equal.

If $d<0$, then the roots are complex conjugates.

Equations of Plane Curves:

Line: $y=mx + b$, where m = slope and $b=y$ intercept

OR

$y-y_1 = m(x-x_1)$, (x_1, y_1) is a point on the line and $m=\text{slope}$

Circle: $(x-h)^2+(y-k)^2=r^2$, where (h,k) is the center and $r=\text{radius}$.

Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where a and b are the major and minor axes.

Parabola: $(x-h)^2=2p(y-k)$, where (h,k) is the vertex and p is the distance from vertex to directrix.

Hyperbola: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where a and b are the major and minor axes.

Logarithms

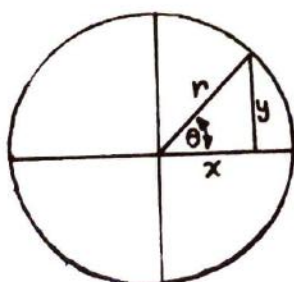
$\log_B N=E$ is equivalent to $B^E=N$.

$\log_c 1=0$ $\log_c c=1$ $\log_c c^b=b$ $\log_c a^b=b \log_c a$

$\log_c \sqrt[b]{a} = (\log_c a)/b$ $\log_c ab = \log_c a + \log_c b$ $\log(a/b) = \log_c a - \log_c b$

$$\log_b a = \frac{\log_c a}{\log_c b}$$

Trigonometry



$$r = \sqrt{x^2 + y^2}$$

SOH-CAH-TOA

$\sin \theta = y/r$ $\csc \theta = r/y$ Amplitude of $n \sin r = n$
 $\cos \theta = x/r$ $\sec \theta = r/x$ Period of $n \sin r = 360/r$
 $\tan \theta = y/x$ $\cot \theta = x/y$

$$\sin \theta = \cos(90-\theta)$$

$$\tan \theta = \cot(90-\theta)$$

$$\sec \theta = \csc(90-\theta)$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\cot^2 \theta + 1 = \csc^2 \theta$$

$$\sin 2A = 2\sin A \cos A \quad \cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

$$\sin(A/2) = \pm \sqrt{\frac{1 - \cos A}{2}} \quad \cos(A/2) = \pm \sqrt{\frac{1 + \cos A}{2}} \quad \tan(A/2) = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}} = \csc A - \cot A$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\tan(A+B) = (\tan A + \tan B) / (1 - \tan A \tan B)$$

$$\tan(A-B) = (\tan A - \tan B) / (1 + \tan A \tan B)$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

Geometry

(A=Angle, S=Side)

Two triangles are congruent if you can prove SSS, SAA or ASA.

Two triangles are similar if you can prove AA.

Two right triangles are congruent if you can prove that their hypotenuses are equal and that one of their legs are equal.

The sum of the angles in a triangle is 180 degrees.

In a right triangle with legs a and b and hypotenuse c, $a^2 + b^2 = c^2$

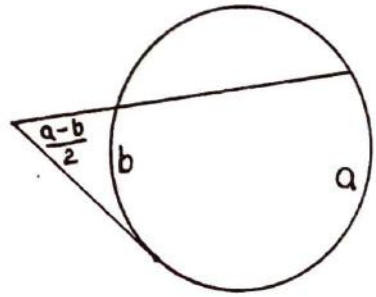
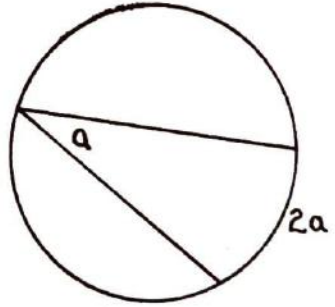
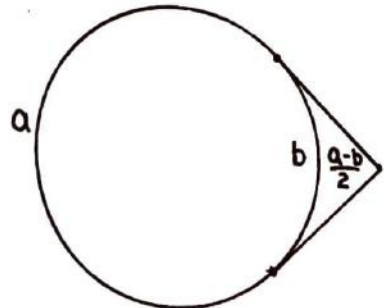
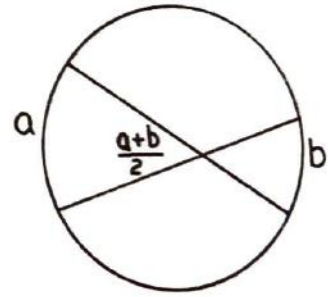
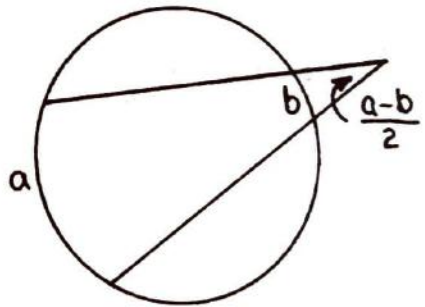
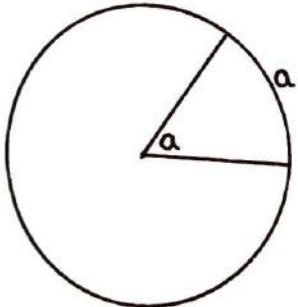
The area of a triangle is $(1/2)bh$ (h=altitude to side of length b).

The area of a trapezoid is $(1/2)(b+B)h$

The area of a parallelogram is bh.

The area of a regular n-gon is $(1/2)ap$ (a=apothem, p=perimeter)

In a circle, the diameter that is perpendicular to a chord bisects the chord.



THE 5 REGULAR POLYHEDRA

by Mark Basch

One of the most intriguing facts in mathematics is the existence of only 5 regular polyhedra. A regular polyhedra is a 3-dimensional figure in which all of the faces are the same regular 2-dimensional shape. The following chart displays the names of all the regular polyhedra, the shapes of the faces which compose it, and the number of faces in each.

To prove that there are only five regular polyhedra we take a regular polyhedron with E edges, V vertices, and F faces each of which is an n -sided regular polygon, having r edges directed out of each vertex. The first theorem we use is Euler's formula which states:

$$(1) V - E + F = 2$$

The next equation we use is

$$(2) nF = 2E$$

Polyhedra	Shape of Face	# of Faces
Tetrahedron	Triangle	4
Cube	Square	6
Octahedron	Triangle	8
Icosahedron	Triangle	20
Dodecahedron	Pentagon	12

This is true because each edge belongs to two faces and is therefore counted twice in nF . The next equation is:

$$(3) nV = 2E$$

This is true because each edge is bounded by two vertices. By substitution into Euler's formula (1) we find

$$(4) \frac{2E}{n} + \frac{2E}{r} - E = 2$$

We then simplify:

$$(5) 1/n + 1/r = 1/2 + 1/E$$

We knew that $n \geq 3$ and that $r \geq 3$ because a polygon must have at least 3 sides and at least 3 edges meet at one vertex. Also, we cannot have both $(n$ and $r) \geq 3$ or else $1/n + 1/r < 1/2$. Hence $1/n + 1/r < 1/2 + 1/E$. If $n=3$ then:

$$(6) 1/r = 1/E + 1/6$$

In this case $r = 3, 4$ or 5 which correspond respectively to a tetrahedron, an octahedron, and an icosahedron. If $r=3$ then:

$$(7) 1/n = 1/E + 1/6$$

In this case $n = 3, 4$ or 5 which correspond to a tetrahedron, a cube and a dodecahedron. Thus these are the only five regular polyhedra!

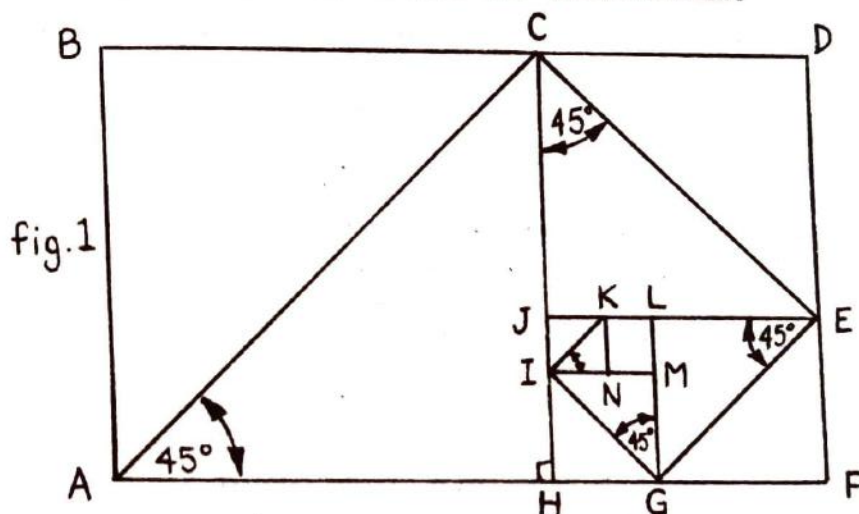
Q.E.D.

A LOGARITHMIC SPIRAL

by Nir Hacoen

From all the peculiar figures of geometry, the Golden Rectangle has some of the most interesting and beautiful properties. Its properties can be expressed in terms of infinite series, quadratic equations, and geometric expressions. A Golden Rectangle has the following basic property: if one cuts away a square from the rectangle, then the rectangle that remains has exactly the same shape (ratio of sides) as the original rectangle. Using this rectangle we can construct the zig-zag spiral. To construct the logarithmic spiral, we must first construct a spiral zig-zag on whose vertices will lie the spiral.

Starting at the lower left of a Golden Rectangle ABDF. (see fig.1), we draw a 45° line to the upper edge at C. From C we draw another 45° line to E on the adjacent side DF. Once again we draw a 45° line from E to G on the adjacent side FA. In addition from each point to which we draw a 45° line, we drop a perpendicular to the opposite side. This process continues ad infinitum.



Each new vertex (such as C, E, G, etc.) belongs to a new rectangle (CDFH, EFHJ, GHJL, etc) which is a 90° turn from the previous one and whose sides are m times those of the previous sides. Each rectangle is rotated 90° and reduced by a factor of m .

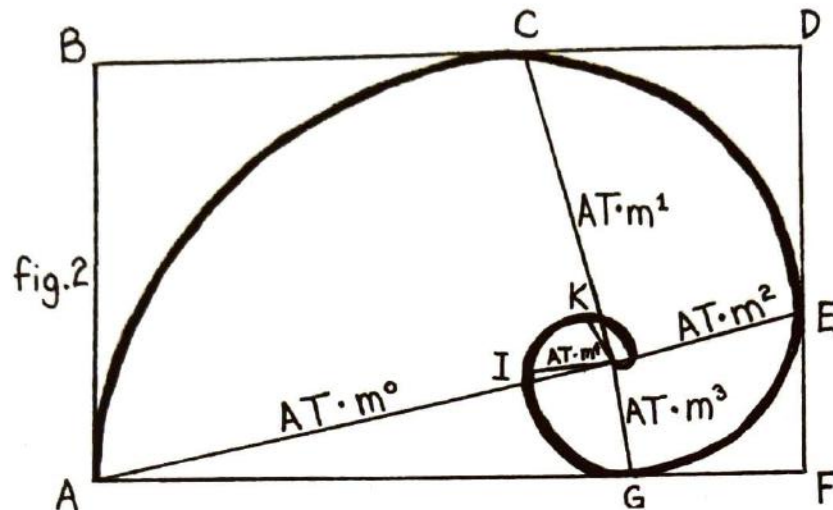
The remarkable property of the figure is that a formula can be derived for its vertices A, E, E, G, I,...

First, pick T (the target point on which the zig-zag closes down) as the origin of the coördinates; let r represent the distance from T to any point on the zig-zag, and choose line AT as the initial direction. Take the $1/4$ turn ($1/4 \cdot 360^\circ = 90^\circ$) as the unit angle and let t denote the number of quarter turns. The vertices A, E, E, G, ... are obtained by rotating AT about T, the origin, through 0, 1, 2, 3, ... quarter turns respectively, and by shrinking the distance AT by $m^0, m^1, m^2, m^3, \dots$ respectively. Therefore, r , the distance from T to any vertex of the spiral zig-zag can be written as follows:

$$r = AT \cdot m^t$$

If we take AT as a unit distance, and thus set $D = r/AT$, we get:

$$D = m^t$$



Substituting all real positive values for t in this equation will give us a continuous spiral.

Interestingly enough, this spiral is not confined purely to mathematics. It manifests itself accurately and beautifully in a snail's shell. The appearance of the spiral is just one more of the countless manifestations of mathematics in our universe. It seems that mathematics is a universal language with which we can represent most phenomena and events.

GUG and GOG

'In a certain tribe, in which polygamy was practiced, a married man's standing in the tribe depended upon the combined weight of his wives- the greater the combined weight, the more important was the man. Every year, on weighing day and according the custom, the married man would stand their wives on neatly spread animal skins, and the chief of the tribe would come around with a crude seesaw and balance the wives of one man against those of another in order to determine the relative importance of the men. Now Gog had only one wife, who was very heavy, while Gug had two much slenderer wives, and all year the two men argued as to who was more important. When weighing day arrived, Gog placed his wife on a large hippopotamus skin, and Gug placed his wives on two small gazelle skins. When weighing was performed, it was found that Gog's wife exactly balanced against the two wives of Gug. Thus it turned out that the two men were equally important, since by the chief's ruling, "the squaw on the hippopotamus is equal to the sum of the squaws on the other two hides."¹

¹ Howard W. Eves. In Mathematical Circles. (Boston:Prindle, Weber, & Schmidt, Inc.) p.8,9.

THE FOURIER SERIES

by Carl Weiner

It is sometimes more convenient to represent a function as an infinite series. A Taylor series, for instance, allows a function to be represented as an infinite polynomial whose terms are equal to $\frac{f^n(0)}{n!} x^n$ (where $f^n(t)$ is the n^{th} derivative of $f(t)$), which is

very useful for approximating values of such functions as $\sin(x)$ and e^x that can be represented as simple Taylor series. A Fourier series represents a function on a given interval as a sum of sine and cosine curves, a form which has many uses in physics, especially in the fields of electronics and sound. One advantage of the Fourier series over Taylor series is that it can represent discontinuous functions.

The technique for finding the Fourier series for a function is similar to the method for deriving a Taylor series. We assume that the function, $f(x)$, is known on the interval $(-\pi, \pi)$ and can be represented in the form

$$f(x) = \left(\frac{1}{2}\right)a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

We can now integrate both sides for $-\pi$ to π

$$f(x)dx = \int_{-\pi}^{\pi} \left[\left(\frac{1}{2}\right)a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right] dx$$

Each term within the summation can be integrated separately, and it is found that

$$\int_{-\pi}^{\pi} a_n \cos nx \, dx = -a_n n (\sin nx) \Big|_{-\pi}^{\pi} = 0$$

$$\int_{-\pi}^{\pi} b_n \sin nx \, dx = b_n n (\cos nx) \Big|_{-\pi}^{\pi} = 0.$$

Therefore,

$$\int_{-\pi}^{\pi} f(x) dx = \int_{-\pi}^{\pi} \left(\frac{1}{2}\right)a_0 = a_0 \pi$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx.$$

Similarly, the function can be multiplied by $\cos nx$ and then integrated:

$f(x) \cos nx = \left(\frac{1}{2}\right)a_0 \cos nx + a_1 \cos x \cos nx + b_1 \sin x \cos nx + \dots + a_n \cos^2 nx + \dots$
 Except for the $a_n \cos^2 nx$ term, all the others are in the form $\sin mx \cos nx$ or $\cos mx \cos nx$ ($m \neq n$). It can be verified that:

$$\begin{aligned} \sin mx \cos nx \, dx &= 0 && \text{in general} \\ \cos mx \cos nx \, dx &= 0 && \text{when } m \neq n. \end{aligned}$$

Hence, when both sides are integrated:

$$\int_{-\pi}^{\pi} f(x) \cos nx \, dx = \int_{-\pi}^{\pi} a_n \cos^2 nx \, dx = a_n \left(x/2 + \sin 2nx/4n \right) \Big|_{-\pi}^{\pi} = a_n \pi$$

Therefore

$$a_n = 1/\pi \int_{-\pi}^{\pi} f(x) \cos nx \, dx.$$

Note that this result is also valid for $n=0$, which was the reason for writing the first term as $1/2(a_0)$.

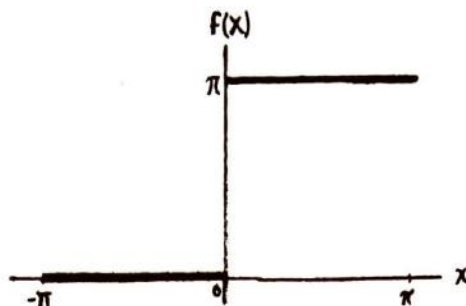
Similarly, the coefficients of the $\sin nx$ terms can be found to be

$$b_n = 1/\pi \int_{-\pi}^{\pi} f(x) \sin nx \, dx.$$

To illustrate the usefulness of the Fourier series, let us find the coefficients a_n and b_n for a function defined by

$$f(x)=0 \text{ if } -\pi \leq x < 0; \quad f(x)=\pi \text{ if } 0 \leq x < \pi$$

which is also called a square wave in electronics, since the function looks as follows:



Applying the equations

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx = \frac{1}{\pi} \left(\int_{-\pi}^0 0 \, dx + \int_0^{\pi} \pi \, dx \right) = \frac{1}{\pi} (\pi^2) = \pi$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{1}{\pi} (\sin nx) \Big|_0^{\pi} = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = -\frac{1}{n} (\cos nx) \Big|_0^{\pi} = \frac{1}{n} (1 - \cos n\pi)$$

The factor $(1 - \cos n\pi)$ assumes the values of 2 and 0 alternately, and so the coefficients of the $\sin nx$ terms become 2, 0, 2/3, 0, 2/5, 0, etc. The entire Fourier series for $f(x)$ becomes

$$f(x) = \frac{1}{2}\pi + 2 \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \frac{\sin 7x}{7} + \dots \right).$$

This way of representing the square wave function has many applications. If $f(x)$ represents a sound wave (pressure versus time), then the series show what frequencies of sound make up the wave. In this case, $\sin x$, also called the fundamental, is the main component, and frequencies an odd number of times greater make up the rest of the tone. If the fundamental has the frequency of middle C, then the Fourier series for $f(x)$ would indicate the following chord:



(This is found by taking alternate harmonics of C. The higher notes will have amplitudes of $1/3$, $1/5$, $1/7$, etc. of the amplitude of the bottom note.)

If some of the higher frequency components of the series are omitted, let's say from $\sin 7x/7$ on, then the resulting function only approximates the original square wave $f(x)$. The graph of

$$y = \frac{1}{2}\pi + 2(\sin x + \sin 3x/3 + \sin 5x/5 + \sin 7x/7)$$

looks as follows:



In many electronic circuits (particularly computers and other digital devices), a square wave is transmitted through a series of components. Usually the circuit cannot transmit the higher frequency parts of the electrical signal due to a phenomenon called capacitance. As a result, a square wave is usually modified in a circuit to come out looking like the picture above. Other electrical signals which have discontinuities like that of the square wave will lose higher frequency components of their Fourier series when transmitted through circuits.

We should note that the solutions for $f(x)$ as a Fourier series found above is only valid on the interval $(-\pi, \pi)$. As it stands, the function will repeat outside these boundaries with a period of 2π , since $\sin(nx+2\pi) = \sin nx$ and $\cos(nx+2\pi) = \cos nx$. If the function is considered on any interval $(-l, l)$, then the Fourier series becomes:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx.$$

This is more general than just considering $f(x)$ from $-\pi$ to π , but the function is still periodic, now with period $2l$. It is conceivable that l could be allowed to approach infinity, thus allowing any function $f(x)$ on the interval $(-\infty, \infty)$ to be represented as a Fourier series. When this is done, however, it is found that terms $a_n \cos nx$ and $b_n \sin nx$ exist for any n , while for a finite series n was only an integer. Therefore the values of a_n and b_n can be represented as a continuous function:

$$a_n + ib_n = \int_{-\infty}^{\infty} f(x) e^{-i2\pi nx} dx$$

($e^{-iu} = \cos u + i \sin u$). The integral on the right side is also written as $T(f)$, called the Fourier transform. In physics, the Fourier transform can be used to find the amplitude of any frequency components making up a function, an operation which the human ear does instantaneously to sound waves but which can take years of work to derive mathematically for irregular functions. In recent times, the computer has helped to solve this equation for functions defined by large data bases of measured points, an operation which would be virtually impossible by hand. Such analysis is useful in fields such as speech analysis, in which recorded sound waves must be analyzed for all their frequency components, in order to allow computers to "hear" speech in the same way as the human ear.

MULTIPLE INTEGRALS

by Joel Hasen

Suppose there is a region in space and $f(x, y, z)$ is a function of x, y , and z that is continuous in the region. If we divide the region into K subregions, then we form the sum

$$S = \sum_{k=1}^n F(x_k, y_k, z_k) \Delta V_k$$

where x_k, y_k , and z_k are the coordinates of a point in the k^{th} subregion and ΔV_k is the volume of the k^{th} subregion. Since $\Delta V_k = \Delta x \Delta y \Delta z$, we form the sum:

$$S = \sum_{k=1}^n F(x_k, y_k, z_k) \Delta z \Delta y \Delta x.$$

We can now write it as an integral:

$$S = \iiint F(x, y, z) dz dy dx.$$

If $F(x, y, z) = 1$, then S is equal to the volume of the region. If $F(x, y, z) = x$, then S is equal to the first moment of the region with respect to the yz -plane. This is used to find the centroid (center of gravity) of the region. If $F(x, y, z)$ is equal to the density at (x, y, z) , then S is equal to the mass of the region. Since we wish to find the volume of the region, we will set $F(x, y, z) = 1$. Hence, the above integral becomes:

$$V = \iiint dz dy dx.$$

We will treat this integral as iterated (repeated) integrals. We will start in the middle and iterate towards the outside:

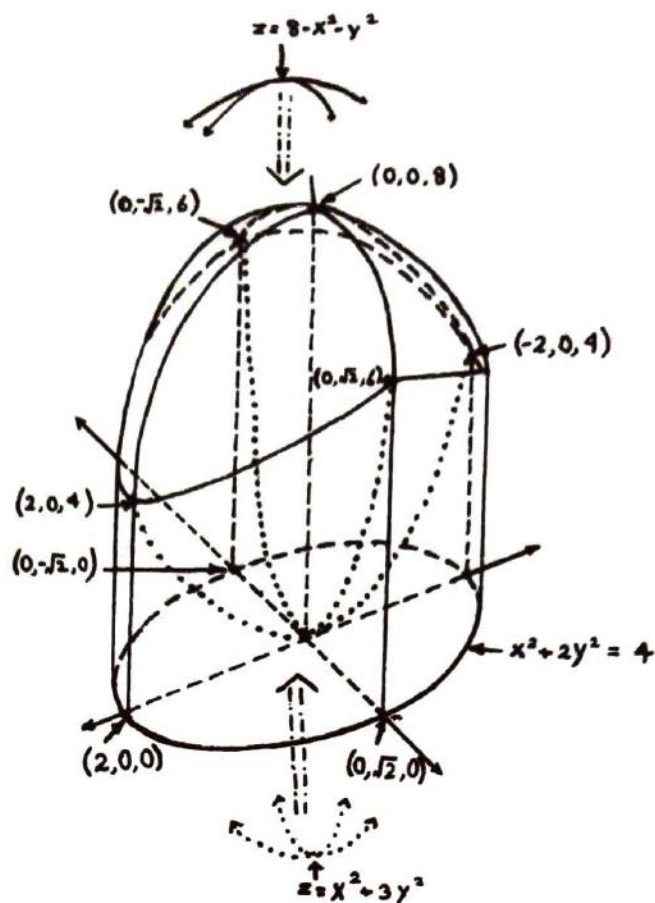
$$V = \int \left[\int \left(\int dz \right) dy \right] dx.$$

The limits of integration of the first integral are $z_1 = f_1(x, y)$ and $z_2 = f_2(x, y)$. During this integration, x and y will be treated as constants. On the second integration, the limits are $y_1 = f_1(x)$ and $y_2 = f_2(x)$. During this integration, x will be treated as a constant. On the last integration, the limits are $x_1 = a$ and $x_2 = b$.

$$V = \int_{c_1}^{c_2} \int_{f_1(x)}^{f_2(x)} \int_{f_1(x,y)}^{f_2(x,y)} dz dy dx.$$

To illustrate this, we may find the volume bounded by the elliptic paraboloids $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

$$f_2(x, y) = z_2 = 8 - x^2 - y^2, \quad f_1(x, y) = z_1 = x^2 + 3y^2.$$



To find $f_2(x)$ and $f_1(x)$, we will find the intersection of the two elliptic paraboloids:

$$\begin{aligned} z &= x^2 + 3y^2 & z &= 8 - x^2 - y^2 \\ x^2 + 3y^2 &= 8 - x^2 - y^2 \\ 2x^2 + 4y^2 &= 8 \\ x^2 + 2y^2 &= 4 \end{aligned}$$

$$y^2 = \frac{4 - x^2}{2}$$

$$y = \pm \sqrt{\frac{4 - x^2}{2}}$$

$$f_2(x) = \sqrt{\frac{4 - x^2}{2}}$$

$$f_1(x) = -\sqrt{\frac{4 - x^2}{2}}$$

To find x_2 and x_1 , again, we will find the intersections:

$$\begin{aligned} \sqrt{\frac{4 - x^2}{2}} &= -\sqrt{\frac{4 - x^2}{2}} \\ 2\left(\sqrt{\frac{4 - x^2}{2}}\right) &= 0 \end{aligned}$$

So $x^2 = 4$, and $x = \pm 2$.

$$x_2 = 2$$

$$x_1 = -2$$

Now we can set up the integral:

$$V = \int_{-2}^2 \int_{-\sqrt{\frac{4-x^2}{2}}}^{\sqrt{\frac{4-x^2}{2}}} \int_{x^2+3y^2}^{8-x^2-y^2} dz dy dx$$

$$dz dy dx$$

$$\begin{aligned}
V &= \iint [z] \frac{8 - x^2 - y^2}{x^2 + 3y^2} dy dx \\
&= \int \int_{-\sqrt{\frac{4-x^2}{2}}}^{\sqrt{\frac{4-x^2}{2}}} \frac{(8 - 2x^2 - 4y^2)}{2} dy dx \\
&= \int_{-2}^2 \left[(8 - 2x^2)y - \frac{4y^3}{3} \right]_{-\sqrt{\frac{4-x^2}{2}}}^{\sqrt{\frac{4-x^2}{2}}} dx \\
&= \int_{-2}^2 2 \left((8 - 2x^2) \left(\frac{4-x^2}{2} \right)^{\frac{1}{2}} - \frac{4 \left(\frac{4-x^2}{2} \right)^{\frac{3}{2}}}{3} \right) dx \\
&= 2 \int_{-2}^2 \left[2(4 - x^2) \frac{(4 - x^2)^{\frac{1}{2}}}{\sqrt{2}} - \frac{4(4 - x^2)^{\frac{3}{2}}}{6\sqrt{2}} \right] dx \\
&= 2 \int_{-2}^2 (4 - x^2)^{\frac{3}{2}} \left(\frac{2\sqrt{2}}{3} \right) dx \\
&= \frac{4\sqrt{2}}{3} \int_{-2}^2 (4 - x^2)^{\frac{3}{2}} dx
\end{aligned}$$

Substitution yields:

$$\begin{aligned}
V &= \frac{4\sqrt{2}}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (4 - 4\sin^2 \theta)^{3/2} (2\cos \theta d\theta) \\
&= \frac{8\sqrt{2}}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (4(1 - \sin^2 \theta))^{3/2} (\cos \theta) d\theta \\
&= \frac{8\sqrt{2}}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (4\cos^2 \theta)^{3/2} \cos \theta d\theta \\
&= \frac{8\sqrt{2}}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 8\cos^3 \theta \cos \theta d\theta \\
&= \frac{64\sqrt{2}}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 \theta d\theta
\end{aligned}$$

$$\begin{aligned}
\text{Let } x &= 2\sin \theta \\
x^2 &= 4\sin^2 \theta
\end{aligned}$$

$$dx = 2\cos \theta d\theta.$$

Since $\theta = \sin^{-1}\left(\frac{x}{2}\right)$, then:

$$\theta_2 = \sin^{-1}\left(\frac{2}{2}\right) = \sin^{-1}(1) = \frac{\pi}{2}$$

$$\theta_1 = \sin^{-1}\left(-\frac{2}{2}\right) = \sin^{-1}(-1) = -\frac{\pi}{2}$$

Since $\cos 2\theta = 2\cos^2 \theta - 1$, then $\cos^2 \theta = (1/2)(\cos 2\theta + 1)$.

$$\begin{aligned}
\cos^4 \theta &= (\cos^2 \theta)^2 = (1/4)(\cos^2 2\theta + 2\cos 2\theta + 1) \\
&= (1/4)\left[(1/2)(\cos 4\theta + 1) + 2\cos 2\theta + 1\right] = \frac{\cos 4\theta}{8} + 1/8 + \frac{\cos 2\theta}{2} + 1/4
\end{aligned}$$

17 Methods of Mathematical Proof

by Romi Goldwasser and Evan Goldberg

If the proof of a theorem is not immediately apparent, it may be because you are trying the wrong approach. Below are seventeen effective methods of proof that may aim you in the right direction.

1. Proof by the Reader's Laziness--"This is easily verifiable by the reader, but consists of endless lines of horribly boring computation."
2. Proof by Change of Coordinates--"We'll rotate the axes until it's true."
3. Proof by Design--If it's not true in today's math, invent a new system in which it is.
4. Proof by Distraction--"Look! Up there!"
5. Proof by Total Obnoxiousness--"Anyone who disagrees with this theorem...SHUT UP!!!"
6. Proof by N-Dimensions--To picture things in the 5th dimension, you need 4th-dimensional drugs. Until these are discovered, it's perfectly safe to talk nonsense in n dimensions, $n > 4$.
7. Proof by Lost Reference--"I know I saw it somewhere..."
8. Proof by Imagination--"Well, we'll pretend it's true..."
9. Proof by Mumbo-Jumbo--" $\forall \alpha \in \Phi, \exists \beta \ni \alpha + \beta = e \dots$ "
10. Proof by Plagiarism--"As we see on page 289..."
11. Proof by Clever Variable Choice--"Let A equal the number such that this proof works."
12. Proof by Necessity--"It had better be true, or the entire structure of mathematics will crumble to the ground."
13. Proof by Simplification--"This proof reduces to the statement $1+1=2$."
14. Proof by Hasty Generalization--"Well, it works for 17, so it works for all reals."
15. Proof by Convenience--"It would be very nice if it were true, so..."
16. Proof by Tessellation--"This proof is the same as the last."
17. Proof by Heptadecation--There's a 17 involved, so it must be true. This method, incidentally, validates all the previous ones.

Compiled by Romi Goldwasser and Evan Goldberg

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$$\cos^4 \theta = \frac{\cos 4\theta}{8} + \frac{\cos 2\theta}{2} + 3/8$$

$$\therefore \int_{-\pi/2}^{\pi/2} \cos^4 \theta \, d\theta = \int_{-\pi/2}^{\pi/2} \left(\frac{\cos 4\theta}{8} + \frac{\cos 2\theta}{2} + 3/8 \right) d\theta$$

$$\text{and } V = \frac{64\sqrt{2}}{3} \left[\frac{\sin 4\theta}{32} + \frac{\sin 2\theta}{4} + \frac{3\theta}{8} \right]_{-\pi/2}^{\pi/2}$$

$$V = \left(\frac{64\sqrt{2}}{3} \right) (2) \left(0 + 0 + \frac{3\pi}{16} \right)$$

Hence, the volume of the region is $8\sqrt{2}\pi$ cubic units.

A PIG FAIR

by Dana Randall

Farmer Brown and Farmer Jones decided that they should enter their pigs in the county fair since they knew they had quite a few prizewinners. All they had to do was ask Farmer Smith, who was running the fair, the size and shape of their plot of land so they could prepare an attractive display for their pigs. Unfortunately, Farmer Smith was also entering some pigs and did not want very much competition. He realized, however, that if the farmers did not have enough time to prepare for the fair they would have a large disadvantage. He thought quickly and told the farmers that the plot of land they were assigned to is in the shape of a right triangle with integral sides where the lengths of the legs are relatively prime and the area of the plot is a perfect square. He assumed they would spend so much time on the problem that there would be no time left to get the pigs ready. Before either of the other farmers said anything, he left.

It was not difficult to see that Farmer Jones was terribly confused by the problem and angry at what had just happened. Farmer Brown, on the other hand, just began scribbling something down on a piece of paper. Before moving to the country he was a major in math at his university, and in a few minutes he too became furious and declared Farmer Smith a liar. He gave the following explanation to his friend:

"If the sides of our plot of land are x , y and z , where z is the hypotenuse, either the legs are both even, both odd, or one of each. They cannot both be even in Farmer Smith's triangle since the legs are relatively prime. If they are both odd then x can equal $(2n+1)$ and y can equal $(2m+1)$. Substituting these values into the Pythagorean Theorem, we get $z^2 = 4n^2 + 4n + 4m^2 + 4m + 1$, which is congruent to 1 when divided by 4. However, a perfect square can only equal 0 or 1 when divided by 4, so we must conclude by contradiction that either x or y , but not both, must be odd. Let x be even.

"If we work with the Pythagorean theorem and the fact that y and consequently z are odd, we can write x , y , and z in terms of the parameters a and b . Since $x^2 = (y+z)(z-y)$ and y and z are both odd, $(y+z)$ and $(z-y)$ are both even. If they have any common divisor besides 2, their sum $(2z)$ and their difference $(2y)$ must also be a multiple of this divisor, and substituting into the pythagorean theorem, $2x$ must also be a multiple of this divisor. But x and y are relatively prime so $(y+z)/2$ and $(z-y)/2$ must be relatively prime. From this information we can deduce that $(y+z)$ and $(z-y)$ are each twice a square, say a^2 and b^2 . The parametric equations we are looking for are therefore $x=2ab$, $y=a^2-b^2$, and $z=a^2+b^2$, where a and b are of opposite parity.

"Now, all we have to do is show the impossibility of the system of equations $x^2+y^2=z^2$ and $y^2+z^2=t^2$, where x , y , z , and t are relatively prime positive integers. We can rewrite the equations as $t^2-x^2=2z^2$ and $t^2-x^2=2y^2$, and we know that y is even and x , z , and

t are odd using the same logic as before. Since t and x both odd, we can write t as $x+2k$ and substitute this value into the above equations. From the first equation we find that $(x+k)^2+k^2=z^2$. From our last results we know there must be two numbers a and b such that $(x+k)=2ab$ and $k=(a^2-b^2)$. Therefore, $t^2-x^2=(t-x)(t+x)=2k(2)(x+k)=8(a^2-b^2)ab$. Substituting into the second equation $(t^2-x^2)=2y^2$, we find $y^2=4ab(a^2-b^2)$. Since a and b are relatively prime, there must be three integers, u , v and w such that $a=u^2$, $b=v^2$ and $a^2-b^2=w^2$. Now, from $w^2=a^2-b^2=(a-b)(a+b)=(u^2-v^2)(u^2+v^2)$ we know that there are two factors of w , m and n , such that $u^2+v^2=m^2$ and $u^2-v^2=n^2$. Since all the variables thus far have been positive integers, we can show that u , v , m and n are all less than t as follows: $t = (x+k) + k = 2ab + (a^2-b^2) > b^2 + (a^2-b^2) = a^2 = u^4 > v^4$ (since $a > b$). Thus u and v are less than t . Since $t > a^2-b^2 > a+b = u^2+v^2 = m^2 > n^2$, m and n are consecutively both less than t , as well. But the equations $u^2+v^2=m^2$ and $u^2-v^2=n^2$ are both of the same form as our original equations $x^2+y^2=z^2$ and $y^2+z^2=t^2$.

This implies that given the above system of equations there will always be a system of equations of the same form, where the variables are continually smaller positive integers. Since there is only a finite number of positive integers less than a given positive integer t and this implies that there are an infinite number of positive integers less than t , we have reached another contradiction and proved the impossibility of the system of equations $x^2+y^2=z^2$ and $y^2+z^2=t^2$.

"Using this proof, we can finally show that the area of Farmer Smith's triangle cannot be a perfect square. If the area, $\frac{1}{2}xy$, is equal to h^2 , then we have the system of equations $x^2+y^2=z^2$ and $xy=2h^2$. Hence, there must be two integers a and b such that $x=2ab$ and $y=a^2-b^2$. Substituting into the second equation, we get $h^2=ab(a-b)(a+b)$. Each of the factors of h^2 must also be relatively prime, and so each must be a perfect square. Let $a=s^2$, $b=r^2$, $a+b=p^2$ and $a-b=q^2$. From this we know that $s^2+r^2=p^2$ and $s^2-r^2=q^2$. Since such a system is impossible, the area of Farmer Smith's triangle cannot be a perfect square, and he must have lied to us."

Farmer Jones, who was confused at the beginning by the word "hypotenuse," did not understand anything his friend had said and just nodded to avoid complete embarrassment. However, since the farmers became very angry at their opponent, they worked twice as hard to beat him, and left the fair with seventeen prizewinning pigs.



TRICKS ARE FOR PIGS

by Ted Poon

Young Judy was furious. She had just memorized her multiplication table from 1 to 9, but her mother said that if she didn't know every product xy , where x and y are integers between 10 and 20, she would be forbidden to play with her yellow pig collection. Judy would be given seventeen days to study. Then she would be given a test of seventeen of these products. If Judy did not answer correctly within seventeen seconds for each question, she would not be allowed to look at her yellow pigs for 289 days.

Even madder now, she started to memorize. She decided to do the perfect squares first. Eleven times eleven equals one hundred twenty-one. Twelve times twelve equals one hundred forty-four. She found this task extremely boring like any English class; however, at the seventeenth product, out of the yellow, she discovered a pattern.

Little Judy noticed that in 17 times 17, we would get the required product if we took $(17 + 7)$ tens and added (7×7) :

$$240 + 49 = 289 = 17 \times 17$$

This was also true for 16×16 :

$$(16 + 6) \times 10 + (6 \times 6) = 256 = 16 \times 16$$

Like the smart girl that she is, she knew that her problem would be solved if this technique generalized. She tried 15×17 . "We have $15 + 7$ or 22 tens which must be added to 5×7 or 35."

$$220 + 35 = 255 = 15 \times 17$$

It works!

Overly impatient, she now wanted to "discover" a proof for this method. "Since the two numbers are between 10 and 20, let one equal $(10 + a)$ and the other equal $(10 + b)$, where $1 \leq (a, b) \leq 9$. My plan is to have $(10 + a + b)$ tens added to ab . We would get

$$(10 + a + b)10 + ab = 100 + 10a + 10b + ab.$$

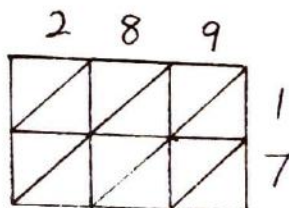
However, by direct multiplication,

$$(10 + a)(10 + b) = 100 + 10a + 10b + ab.$$

I have proved it!"

Well, Judy now knew that she would pass her test. However, since she discovered the trick in one day and had lots of time left, she wanted to find new tricks in the 16 days that followed.

Principally, she wanted to think of tricks to be done mentally, but she saw nothing wrong with tricks on paper. It took her a whole week to see her next trick, lattice multiplication. Suppose you have the problem 289×17 . Instead of just direct multiplication involving carrying of numbers, we can complete the problem by using a lattice as shown



We now multiply one-digit numbers two at a time and place the product in the correct position.

	2	8	9	
	2	8	9	1
1	4	5	6	7
	4	6	3	

Next, we just add along the diagonals, which does involve some carrying, but this carrying is less demanding than carrying done at the same instant as multiplying.

	2	8	9	
4	1	5	6	
	4	6	3	
	9	1	3	

The correct answer is 4913.

Incidentally, Judy found out only one more trick before the 17 days. This one involved squaring numbers ending in 5:

$$(10x + 5)^2 = 100x(x+1) + 25.$$

For example, to square 65:

$$6 \cdot 7 = 42 \text{ and adding 25 yields } 4225 = 65^2.$$

Greatly improved Judy now had to take her mother's test. Her mother almost fainted when Judy finished the seventeen questions in less than two minutes. So impressed was Mrs. Riemann that she decided to buy Judy 17 more yellow pigs.

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MEDITATING ON TRANSCENDENCY

by Iuval Clejan & Anna Franco

What are transcendental numbers? This question, which appears to be the first step in understanding the meaning of the Universe, has a surprisingly simple answer. By definition, transcendental numbers are complex numbers that are not the roots of any polynomial with integer coefficients. It can be proved that numbers such as $e, \pi, e^\pi, \sum_{j=1}^{\infty} (10)^{-j!}$ and others are transcendental. But we will content ourselves by proving the existence of such numbers. First let us define a few terms.

Definition: An algebraic number is a number which is the root of a polynomial with integer coefficients.

Definition: A set is countable if its elements can be listed in an infinite sequence.

Definition: The index of the polynomial $a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, where $a_n \neq 0$, is equal to $n + |a_n| + |a_{n-1}| + \dots + |a_0|$, where the a_i 's are integers.

Theorem 1 : The set of real numbers between 0 and 1 is not countable.

Proof: Assume it is countable. Then it could be listed in the sequence

$$\begin{aligned} a_1, a_2, a_3, \dots, \text{ where } \begin{aligned} a_1 &= 0.a_{11}a_{12}a_{13}a_{14}\dots \\ a_2 &= 0.a_{21}a_{22}a_{23}a_{24}\dots \\ a_3 &= 0.a_{31}a_{32}a_{33}a_{34}\dots \\ &\vdots \\ &\vdots \\ &\vdots \end{aligned} \end{aligned}$$

etc.

and each line is the decimal representation of a number.

But let us consider $b = 0.b_1b_2b_3\dots$ with $b_i \neq a_{ii}$. This differs from a_i in the i th spot, so $b \neq a_i$ for any positive integer i . Thus the sequence $\{a_i\}$ does not include all the reals between 0 and 1. This contradiction proves the theorem. From this it is also seen that the set of all reals and the set of all complex numbers (which contains the reals) are not countable.

Theorem 2 : The set of algebraic numbers is countable.

Proof: It is easy to see that each positive integer m can be partitioned into the sum of positive integers in a finite number of ways. For example, $3 = 1+1+1 = 2+1$ can be partitioned in two different ways. If we can also show that each partition corresponds to a finite number of polynomials, then we will have proved that each index corresponds to a finite number of polynomials.

For any partition $p_0 + p_1 + p_2 + \dots + p_r = m$, let $a_i = \frac{+}{-}(p_i - 1)$ for $0 \leq i \leq r-1$, and $a_r = \frac{+}{-}(p_r)$. Note that the index of the polynomial $a_r x^r + a_{r-1} x^{r-1} + a_{r-2} x^{r-2} + \dots + a_0$ is $r + |a_r| + |a_{r-1}| + \dots + |a_0| =$

$$= r + p_r + (p_{r-1} - 1) + (p_{r-2} - 1) + \dots + (p_0 - 1) = r + \sum_{i=0}^r p_i - r = m.$$

Now we must show that this method of assigning a finite number of polynomials to each index m includes all polynomials of index m .

NEW YORK UNIVERSITY
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PRESENTS:

.....SOME BRAIN TEASERS TO TEST YOUR METTLE. A SPLENDID AWARD WILL BE
MADE FOR THE BEST SOLUTIONS. GOOD LUCK!!

1. Let "cutting off the corners" of a convex polygon mean the following. On each side of the polygon, we mark the points halfway between the middle of the side and the two endpoints. Then we connect the two marked points closest to each vertex on the two sides meeting at the vertex and remove the triangles thus formed.

Start with the square $|x| < 1$ and $|y| < 1$. If we cut off corners infinitely we get a curve in the limit whose equation is fairly simple. Find the equation.

Show also that if instead of marking the points halfway, we mark the points two-thirds of the way to the endpoints, we get a curve which does not have a continuous second derivative in any interval.

2. A tournament T is a directed graph such that if P and Q are any two nodes of T , then exactly one of $\vec{PQ}$ and $\vec{QP}$ is an edge of T .

A spanning path of a graph is a path $P_1P_2\dots P_n$ in which each node occurs exactly once. Note the first and the last node are not the same.

Show that every tournament with 100 nodes has a spanning path $P_1P_2\dots P_{100}$ such that P_1P_{100} is also an edge of the graph.

3. Applying the Σ operator to a sequence $a_1, a_2, a_3, \dots$ means forming the new sequence $a_1, a_1+a_2, a_1+a_2+a_3, \dots$.

Let k be any positive integer ≥ 2 . Take the sequence of positive integers. Delete every k -th element, apply the summation operator, delete every $k-1$ st element, apply the summation operator again, etc. until the summation operator has been applied $k-1$ times. Prove that the final sequence is $1, 2^k, 3^k, 4^k, \dots$.

Send your solutions to: DEPARTMENT OF MATHEMATICS c/o Liz Kaczmarczyk
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